

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
3. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) `choice(seq)`
 - (b) `randrange ([start,] stop [,step])`
 - (c) `random()`
 - (d) `seed([x])`
5. In python 3, which of the following is floor division?
 - (a) `/`
 - (b) `//`
 - (c) `%`
 - (d) `|`

True / False (10 marks)

Answer True or False

6. True or False: The method 'sort()' can be directly called on a string to sort its characters in Python.
7. True or False: In Python 3, the max function returns the largest value in a list, so `max([1, 2, 3, 4])` equals 4.
8. True or False: In Python 3, the element at index 1 of the list ['a', 'Chemistry', 0, 1] is 'a'.
9. True or False: In Python 3, the `strip([chars])` function removes all leading and trailing whitespace from a string.
10. True or False: $O(n^2)$ represents a larger growth rate than $O(n)$ in terms of asymptotic complexity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain the process of converting the decimal number 231 to its hexadecimal representation. Discuss the significance of this conversion in computer science and provide the final result.
12. Discuss the characteristics of phishing attacks and analyze why a legitimate email from a bank requesting a phone call for transaction verification is least indicative of such an attack.

End of Paper